

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Seventh Semester of B. Tech. (CL) Examination****March-April 2018****CL 406.01 Design of Structures-II****Date: 02.04.2018, Monday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.
5. IS 456:2000, IS 800:2007, IS 875:1987 (Part I-V), SP 16, SP 6, SP 34 and steel table are permitted, provided that they do not contain anything other than the printed matter inside.
6. Assume Fe 415, M 20 and Fe 410 for structural component unless otherwise mentioned.

SECTION – I

Q-1 Design a dog legged stair case for a residential building hall measuring 2.5 m x 5 m. The width of the landing is 1.2 m. The distance between floor to floor is 3 m. The rise and tread may be taken as 150 mm and 250 mm respectively. The live load is 2.5 kN/m². The materials used are M 20 grade concrete and Fe 415 grade steel. Sketch the details of steel. The flight and landing slabs spans in the same direction. **[07]**

Q-2 (A) Answer the followings: **[04]**

a) An R.C.C. beam not provided with shear reinforcement may develop cracks in its bottom in

- | | |
|------------------|-----------------|
| (i) 25° | (ii) 35° |
| (iii) 45° | (iv) 55° |

b) The percentage reinforcement in case of slabs, when high strength deformed bars are used is not less than

- | | |
|-------------------|------------------|
| (i) 0.12 | (ii) 0.15 |
| (iii) 0.20 | (iv) 0.18 |

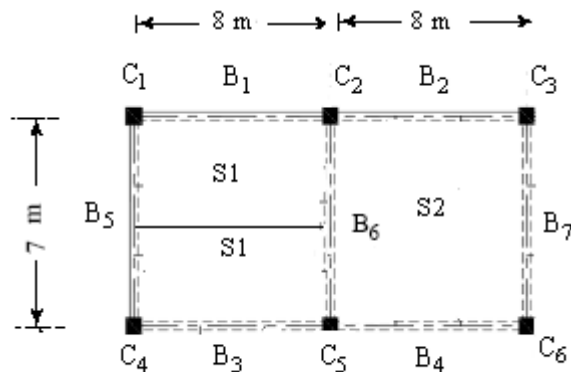
c) The maximum area of tension reinforcement in beams shall not exceed

- | | |
|------------------|------------------|
| (i) 0.15% | (ii) 4% |
| (iii) 2% | (iv) 1.5% |

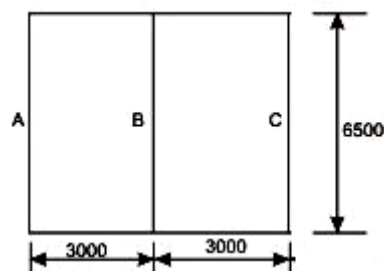
d) The stem of a cantilever retaining wall which retains earth level with top is 6 m. If the angle of repose and weight of the soil per cubic metre are 30° and 2000 kg respectively, the effective width of the stem at the bottom, is

- | | |
|-------------------|------------------|
| (i) 51.5 | (ii) 53.5 |
| (iii) 52.5 | (iv) 54.5 |

- (B) **Figure 2** shows floor plan of a RC beam-column slab having 250 mm × 400 mm beams, 230 mm thick brick masonry walls and 500 mm × 500 mm columns. Live load and floor finish for floor level is 4 kN/m² and 1 kN/m² respectively. Floor to floor height is 3.35 m. Calculate total load on beams B₁, B₂, B₃, B₄, B₅, B₆, and B₇. The span dimensions shown in the figure are effective spans. [10]

**Figure 2****OR**

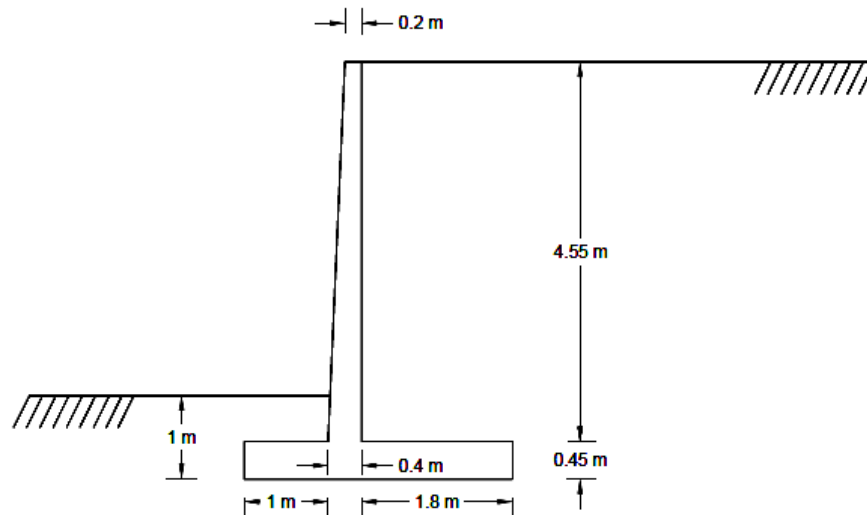
- Q-2 (A) Explain the share of loads by the supporting beams in one-way and two-way slab by suitable drawing. [04]
- (B) Design and detail the one-way continuous slab shown in **Figure 3** subjected to uniformly distributed imposed load of 4 kN/m² using M 20 grade concrete and Fe 415 steel. The load of floor finish is 1 kN/m². The span dimensions shown in the figure are effective spans. The width of beam at the support is 300 mm. [10]

**Figure 3**

(All Dimensions are in mm.)

- Q-3 (A) Determine the plan dimensions of a combined footing for two axially loaded columns C₁ and C₂ having size 300 mm × 300 mm. The column C₁ is flushed with the property line and center to center distance between both the column is 3 m. Column C₁ and C₂ carries factored load of 650 kN and 800 kN respectively. Safe bearing capacity of sub soil is 180 kN/m². [04]
- (B) a) Why shear key is provided in retaining wall? [02]
- b) Draw qualitative deflection shape of cantilever retaining wall. [03]

- c) Design the stem of a cantilever retaining wall as shown in **Figure 4**. The [05]
 density of soil is 18 kN/m^3 and its angle of repose is 30° . The embankment is horizontal at top. The safe bearing capacity of soil is 150 kN/m^2 and the coefficient of friction between the soil and concrete is 0.6. Adopt M 20 grade concrete and Fe 415 steel.

**Figure 4****OR**

- (B) Calculate the factor of safety against sliding and overturning for the counterfort [10]
 retaining wall for the following data:
 Height of wall above GL: 6 m
 SBC of soil: 180 kN/m^2
 Angle of internal friction: 30°
 Density of soil: 16 kN/m^3
 Spacing of counter forts: 3 m c/c
 Materials: M 20 grade concrete and Fe 415 steel
 Also calculate the maximum and minimum soil pressure underneath the footing.

SECTION – II**Q-4 Answer the Followings:****[07]**

- (A) When the bolts are subjected to reversal of stresses, the most suitable type of bolt is
- | | |
|------------------------------|-------------------------------|
| (i) Black bolt | (ii) Ordinary unfinished bolt |
| (iii) Turned and fitted bolt | (iv) High strength bolt |
- (B) If the thickness of a structural member is small as compared to its length and width, it is classified as
- | | |
|-------------------------|----------------------|
| (i) One dimensional | (ii) Two dimensional |
| (iii) Three dimensional | (iv) None of these |

- (C) The effective length of a fillet weld should not be less than
 (i) Two times the weld size (ii) Four times the weld size
 (iii) Six times the weld size (iv) Weld size
- (D) A fillet weld whose axis is parallel to the direction of the applied load, is known as
 (i) Diagonal filler weld (ii) End fillet weld
 (iii) Side fillet weld (iv) All the these
- (E) The best arrangement to provide unified behavior in built up steel columns is by
 (i) Lacing (ii) Battening
 (iii) Tie plates (iv) Perforated cover plates
- (F) The distance between c.g. of compression and c.g. of tension flanges of a plate girder, is known as
 (i) Overall depth (ii) Clear depth
 (iii) Effective depth (iv) None of these
- (G) Factor of safety is the ratio of
 (i) Yield stress to working stress (ii) Tensile stress to working stress
 (iii) Compressive stress to working stress (iv) Bearing stress to working stress

- Q-5 (A) Calculate nodal dead load, live load and wind load for roof truss of an industrial building as shown in **Figure 5** with 16 m span and 48 m long. The roofing is asbestos cement sheets (Dead weight = 171 N/m^2). Spacing between two consecutive trusses is 8 m. The basic wind speed is 47 m/s and terrain category 3 of class B building. The building clear height at the eaves is 11 m. Assume weight of bracing = 12 N/m^2 . [07]

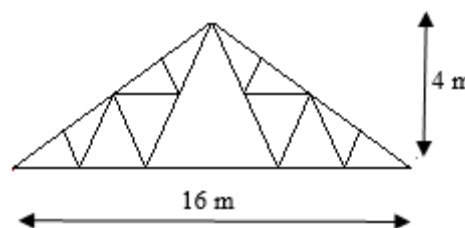


Figure 5

- (B) A gantry girder is to be used in an industrial building carrying a manually operated overhead travelling crane for the following data: [07]
 Minimum clearance between crane hook and the gantry girder: 1.2 m
 Crane capacity: 300 kN
 Weight of the crane girder excluding crab: 300 kN
 Crab weight: 40 kN
 c/c distance between gantry rails: 16 m
 Span of gantry girder: 8 m

c/c distance between the wheel: 3.5 m

Gantry rail self-weight: 300 N/m

Diameter of wheel: 150 mm

Life span in years: 100

Grade of steel: Fe 410

Determine the maximum shear force and bending moment for the moving loads.

OR

Q-5 (A) Design an unequal single angle purlin if spacing of truss is 5 m and spacing of purlin is 1.275 m. Dead load, live load and wind load are 0.21 kN/m², 0.724 kN/m² and 2.586 kN/m² respectively. Use load combination DL+WL. Consider inclination of the roof with the horizontal (θ) = 11.3°. **[05]**

(B) A gantry girder section ISWB 600 @ 1311.6 N/m with ISMC 300 @ 351.2 N/m on its top flange is used in an industrial building. The girder is subjected to following moments and shear force. **[09]**

Transverse moment M_z : 508.21 kNm

Lateral moment M_y : 21.97 kNm

Transverse shear force V_x : 415.07 kN

Grade of steel: Fe 410

Type of Crane: MOT

Check whether ISWB 500 with ISLC 350 on compression flange is adequate to carry moment and shear force. Do all necessary checks.

Q-6 (A) Design a slab base for a column ISHB 300 @ 710.2 N/m subjected to a factored axial compressive load of 1500 kN. The base rests on concrete pedestal of grade M 25. Use Fe 410 grade of steel. **[06]**

Q-6 (B) A welded plate girder section with flange plates 560 mm x 50 mm thick and web plate 1800 mm deep x 12 mm thick is simply supported. The span is 24 m and laterally restrained throughout. It carries a live load of 100 kN/m throughout the span exclusive of self-weight. Check moment and shear capacity of plate girder section. Use Fe 410 grade of steel. **[08]**

OR

(B) Answer the followings:

a) Give general requirement for outstand of web stiffeners as per IS 800:2007. **[02]**

b) Draw bending stress and shear stress diagram of plate girder. **[02]**

c) Define plastic, compact, semi compact & slender section with moment - rotation relationship graph. **[04]**
